Shape and Space 4 — Sine and cosine ratios · Calculating sides and angles

Calculator allowed. Give answers correct to 3 significant figures (angles to 1 decimal place where asked). Draw and label a right-angled triangle for each part. Marks in brackets. ★ = harder question.

1 A window cleaner's ladder is 6 m long and makes an angle of $70°$ with the level ground as it leans against a vertical wall. [4]

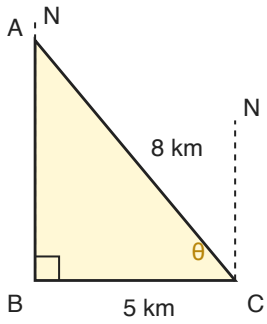
a Calculate the height of the top of the ladder above the ground.

b The cleaner climbs 2.5 m up the ladder. How far is his lower foot from the wall?

a Height is opposite the $70°$ angle, hypotenuse 6: $h = 6\sin 70° = 5.64$ m.

b Foot of the ladder from the wall: $6\cos 70° = 2.052$ m. A point 2.5 m up the ladder is $2.5\cos 70° = 0.855$ m closer to the wall: $2.052 - 0.855 = 1.20$ m.

2 Three phone masts A, B and C: C is 5 km due east of B; A is due north of B and 8 km from C. [8]



a Calculate, to 1 decimal place: **(i)** the angle θ at C, **(ii)** the bearing of A from C, **(iii)** the bearing of C from A.

b Calculate the distance AB in km.

a (i) At C the adjacent side is $BC = 5$ and the hypotenuse is $AC = 8$:
$\cos\theta = \dfrac{5}{8} \Rightarrow \theta = 51.3°$.

(ii) From the north line at C, CA lies $90 - 51.3 = 38.7°$ anticlockwise (towards the north-west), so the bearing is $360 - 38.7 = 321.3°$.

(iii) Reverse bearing: $321.3 - 180 = 141.3°$.

b AB is opposite θ: $AB = 8\sin 51.3° = 6.24$ km (or Pythagoras: $\sqrt{64 - 25} = \sqrt{39} = 6.24$ ✓).

3 A circle with centre O represents the Earth (radius $OW = 6400$ km). A satellite S is at a distance $OS = 42\,000$ km from the centre. W (Washington) and D (Delhi) are points on the circle such that the line OS crosses the chord WD at right angles at N, and angle $WOD = 100°$. [13]

a Calculate, to 3 s.f. or to the nearest degree: **(i)** WN, **(ii)** ON, **(iii)** NS, **(iv)** angle ODS.

b Explain briefly why, if angle ODS were less than $90°$, a signal from the satellite could not reach Delhi.

c A radio signal is sent from W to D via S. Calculate the distance it travels.

d Radio signals travel at $300\,000$ km/s. How long does the signal take?

ON bisects the $100°$ angle (symmetry): angle $WON = 50°$, triangle OWN is right-angled at N with hypotenuse 6400.

a (i) $WN = 6400 \sin 50° = 4903$ km ≈ 4900 km. **(ii)** $ON = 6400 \cos 50° = 4114$ km ≈ 4110 km. **(iii)** $NS = 42\,000 - 4114 = 37\,886 \approx 37\,900$ km.

(iv) In right-angled triangle DNS: $\tan(\angle DSN) = \dfrac{4903}{37\,886} \Rightarrow \angle DSN = 7.4°$. In triangle ODS: $\angle ODS = 180 - 50 - 7.4 = 122.6° \approx 123°$.

b At exactly $90°$ the line DS would be a tangent — grazing the Earth. Less than $90°$ means DS would have to pass *through* the Earth, so D would be over the horizon from S.

c $DS = \sqrt{4903^2 + 37\,886^2} = 38\,200$ km (3 s.f.); by symmetry WS is the same: total $2 \times 38\,200 = 76\,400$ km.

d $\dfrac{76\,400}{300\,000} = 0.255$ s — about a quarter of a second.

Total: 25 marks

Check: SOH CAH TOA — pick the ratio that uses the side you know and the side you want · hypotenuse is always opposite the right angle · angles from two sides: use $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ · calculator in degrees, keep full values between steps.